



Lab 03

COMP1323 Networks and Security

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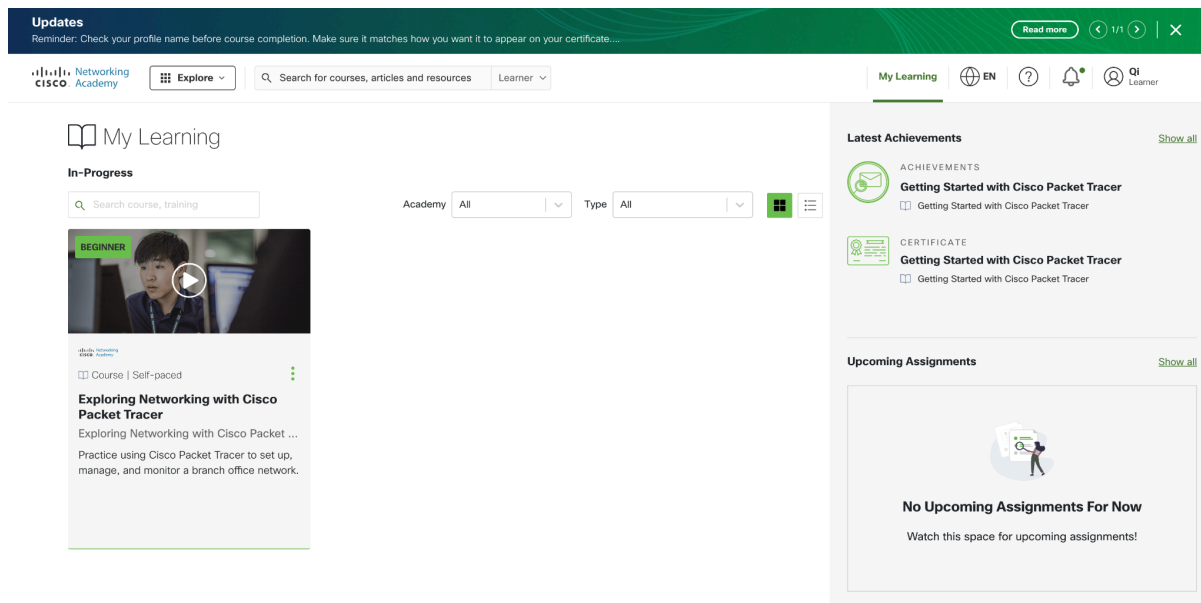
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Date: March 7, 2026

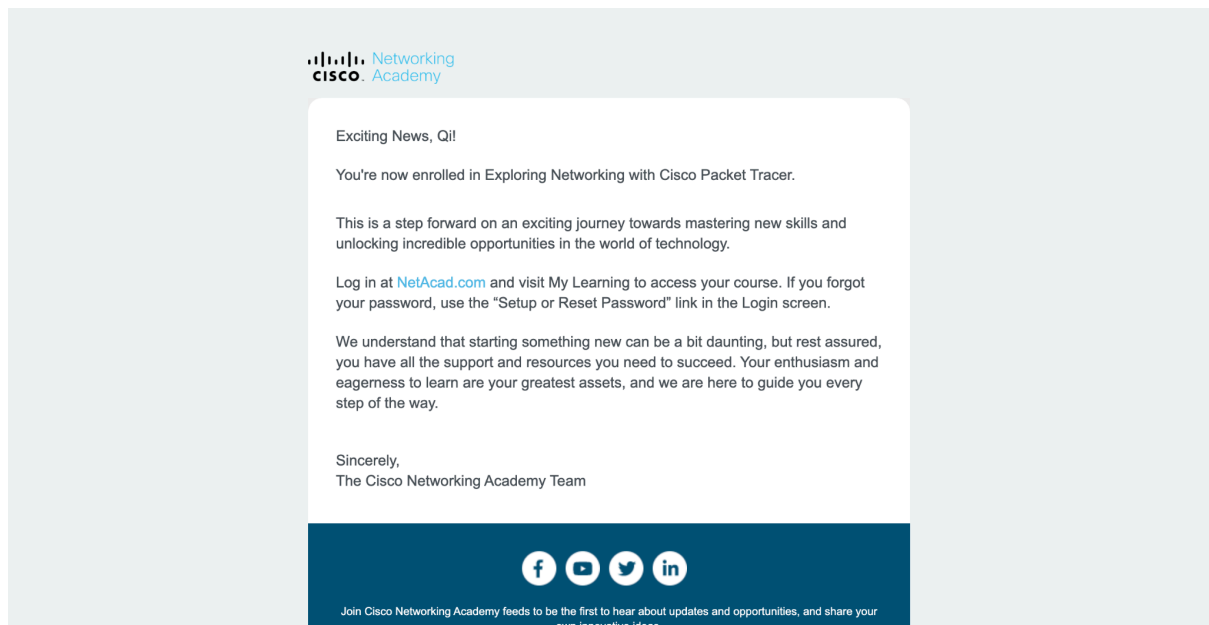
Instructor: Dr. Syed Hamid Hussain Madni

Activity 1:

- Screenshot of course enrollment



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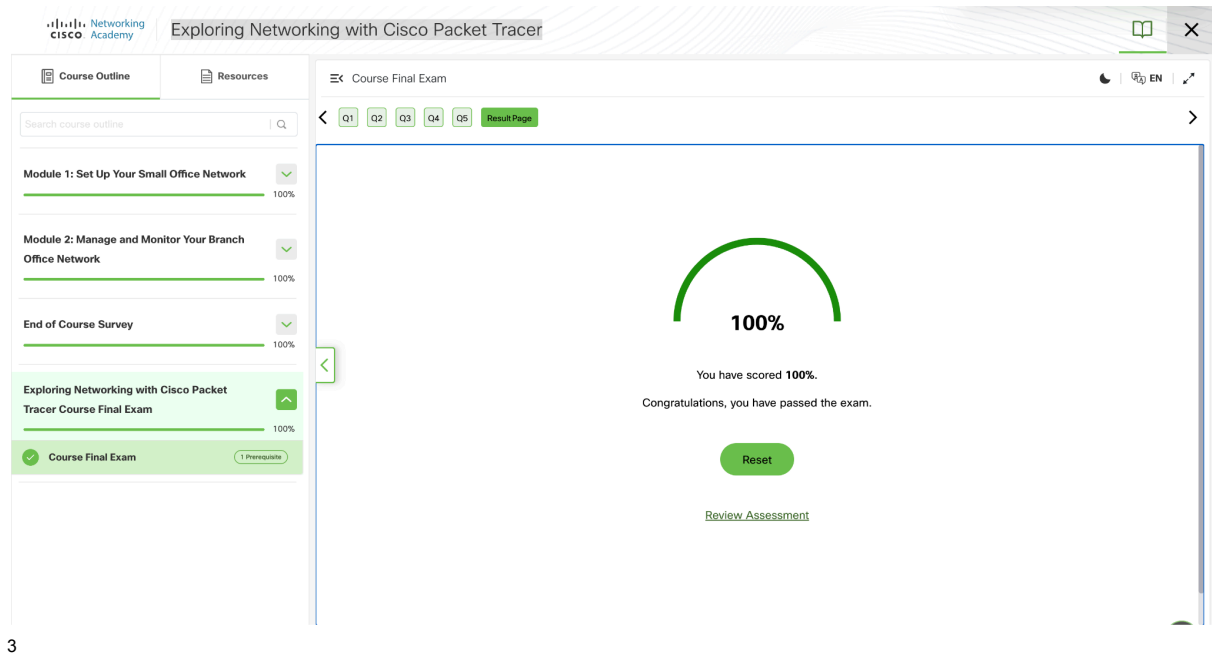


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¹ First screenshot of course enrollment

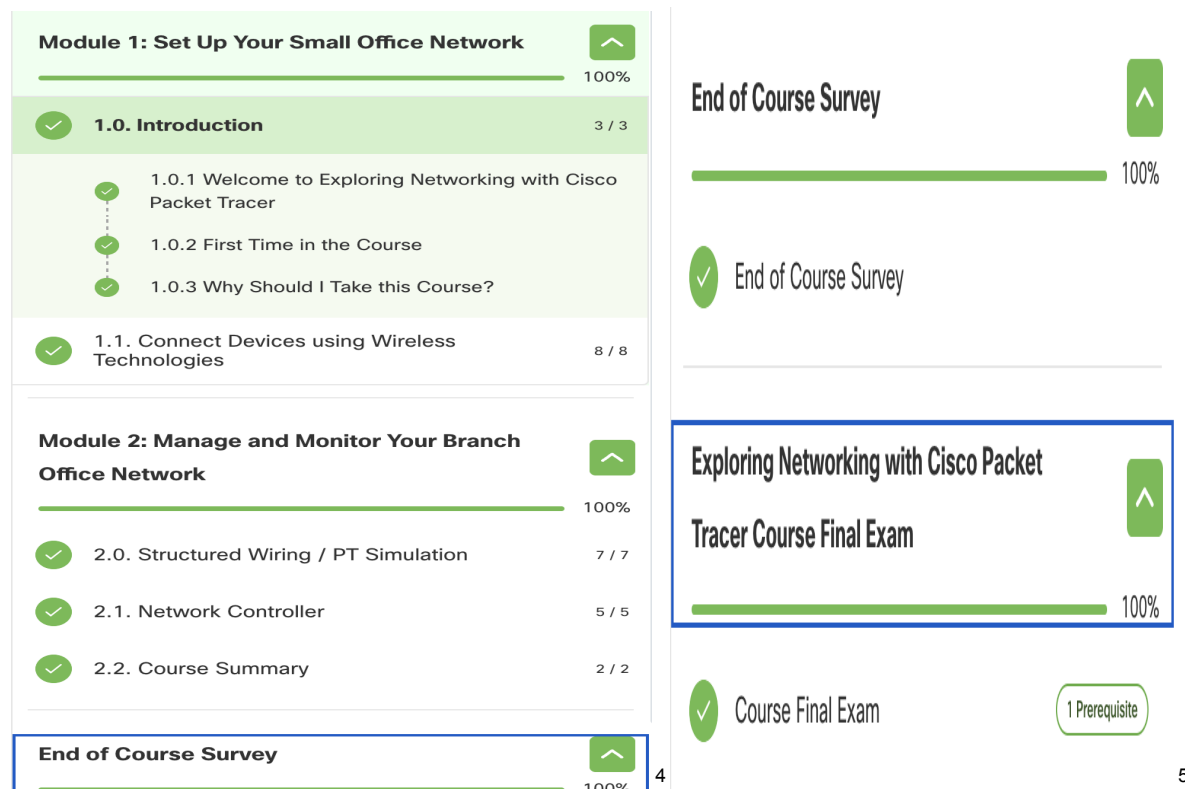
² Second screenshot of course enrollment

- Screenshot showing course progress



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- Screenshot of each completed module



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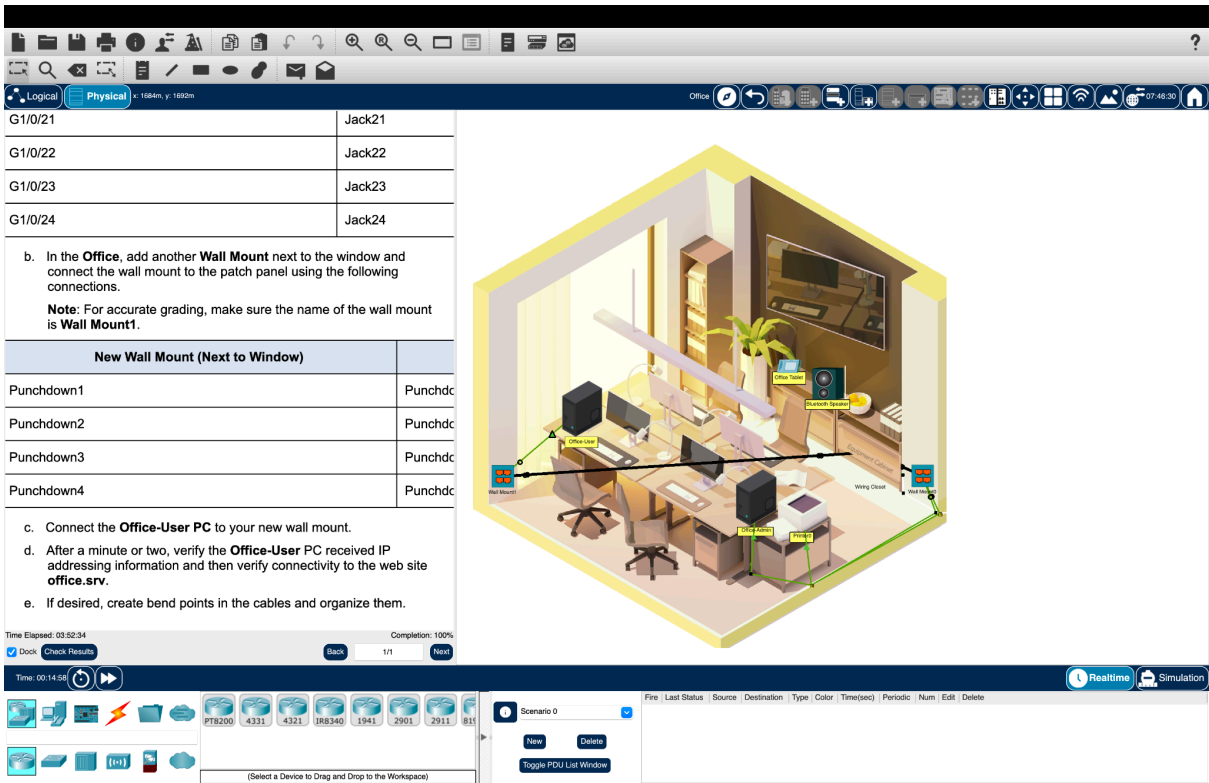
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³ Screenshot showing course progress

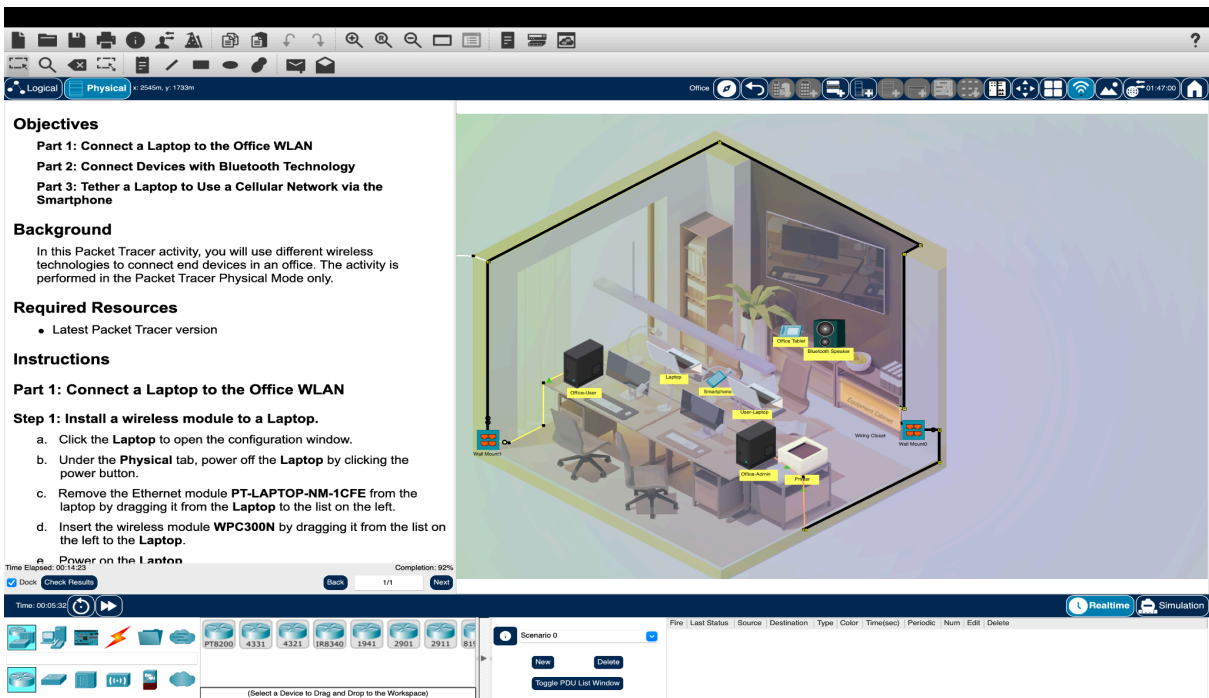
⁴ Screenshot of 1.0 & 2.0 module

⁵ Screenshot of Survey & Final Exam module

• Screenshot of completed tasks/activities inside the modules



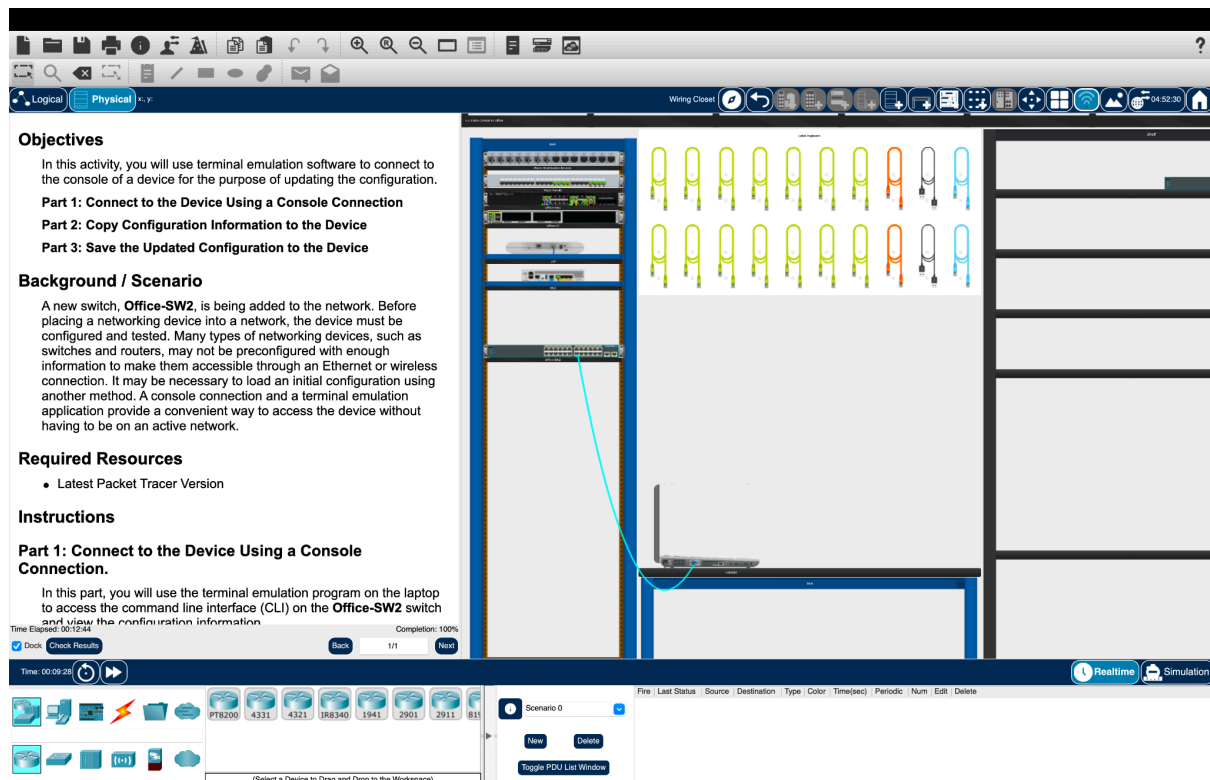
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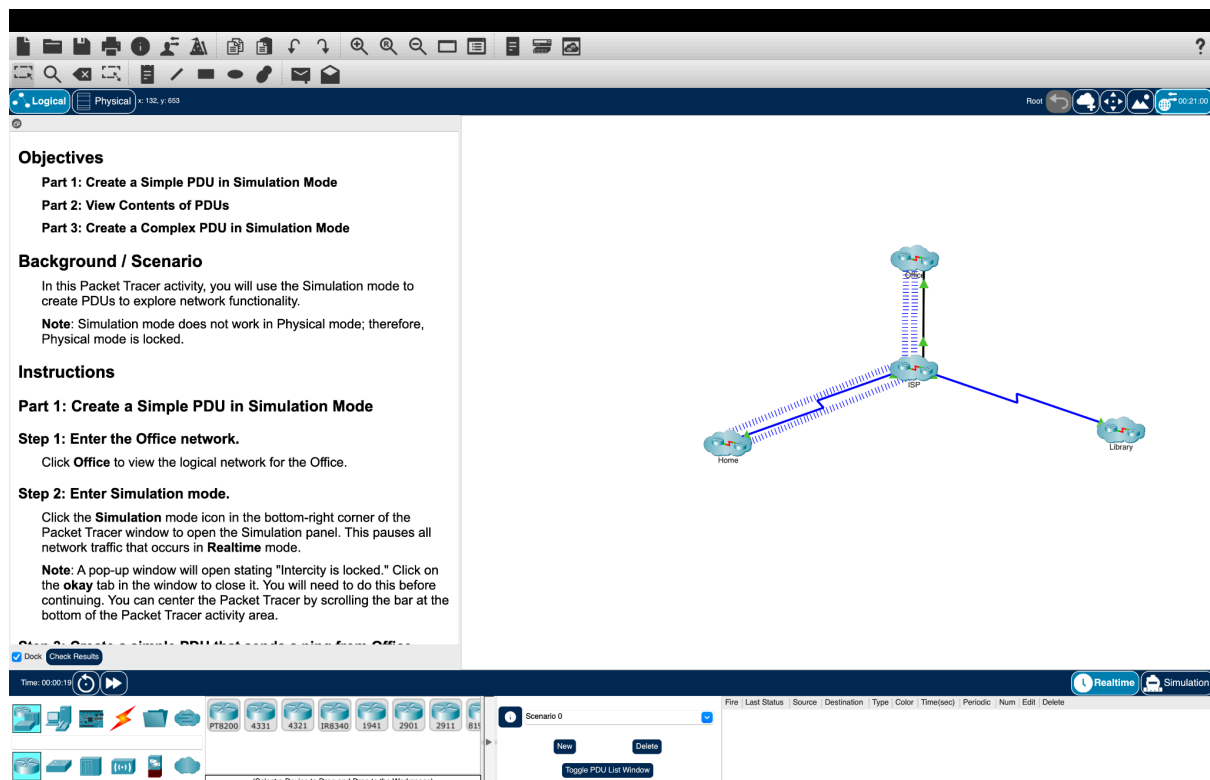
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⁶ Screenshot of 1.1.4 Packet Tracer - Create Realistic Structured Cabling in the Physical Workspace and Cabling Devices in a Rack

⁷ Screenshot of 1.1.6 Packet Tracer - Connect Devices using Wireless Technologies



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⁸ Screenshot of 1.1.8 Packet Tracer - Explore Device Configuration Using the CLI (console)

⁹ Screenshot of 2.0.3 Packet Tracer - Examine Packets in the Small Office

Objectives

In this activity, you will install a patch panel and a wall mount. You will then use these to connect network devices in the office to the equipment in the wiring closet.

Part 1: Add an Additional Switch to the Rack in the Office Network

Part 2: Connect a PC to the Office Network

Part 3: Work with Clusters

Part 4: Add a Second Home Cluster to the Network

Required Resources

- Latest Packet Tracer version

Instructions

Part 1: Add an Additional Switch to the Rack in the Office Network

- In the **Wiring Closet**, click and drag switch **Office-SW2** from the **Shelf** on the right to the **Rack** on the left. It can be placed on any open space on the **Rack**.
- Select a copper straight-through cable from the **Cable Pegboard**.
- Connect the cable to switchport **FastEthernet 0/1** on switch **Office-SW2** and connect the other end to **Jack 1** on the **Patch Panel**.
- Select another copper straight-through cable from the **Cable Pegboard**.
- Connect the cable to **Switchport GigabitEthernet 0/1** on switch **Office-SW2** and connect the other end to **Jack 2** on the **Patch Panel**.

Time Elapsed: 01:46:07 Completion: 100%
☒ Done ☐ Check Results 1/1

Time: 00:10:50

Scenario 0

(Select a Device to Drag and Drop to the Workspace)

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Part 2: Connect a PC to the Office Network

- Click **Place Note** to document the network devices listed from the **NETWORK DEVICE** screen. Place the note in the white space next to the Office background. For example, this info should be part of the note:
Office-SW1: 192.168.20.4

Step 3: Discover new devices in the Office network.

In this step, you will connect a wireless devices to the Office wireless network and observe the changes in the hosts listing.

- From the web browser for **Office-Admin**, navigate to the **Network Controller** if necessary.
- Navigate to the **HOSTS** screen. (Click **Dashboard menu > Assurance > HOSTS**). Note the host list.
- Connect the **Office Tablet** and **Smartphone** to the wireless network. For each device, click the **Config** tab, select **Wireless0** in the left pane, and select **ON** for the Port Status. Verify the devices receive an IP address from the DHCP server. The IP addresses will be from the 192.168.2.x /24 network, where x is ranged between 11 to 254. You can click **Fast Forward Time (Alt + D)** to speed up the process.
- On the Network Controller, navigate to the **DISCOVERY** screen. (Click **Dashboard menu > Provisioning > DISCOVERY**) Select the already configured Discovery process named **Office LAN**. Click **START** to run the discovery process. When the discovery process is finished, the newly connected **Office-Tablet** and **Smartphone** are displayed in the Discovered Devices list.
- Return to the **HOSTS** screen. Note the newly discovered hosts.
What IP addresses are assigned to **Office-Tablet** and **Smartphone**?

Time Elapsed: 00:59:44 Completion: 100%
☒ Done ☐ Check Results 1/1

Time: 00:03:50

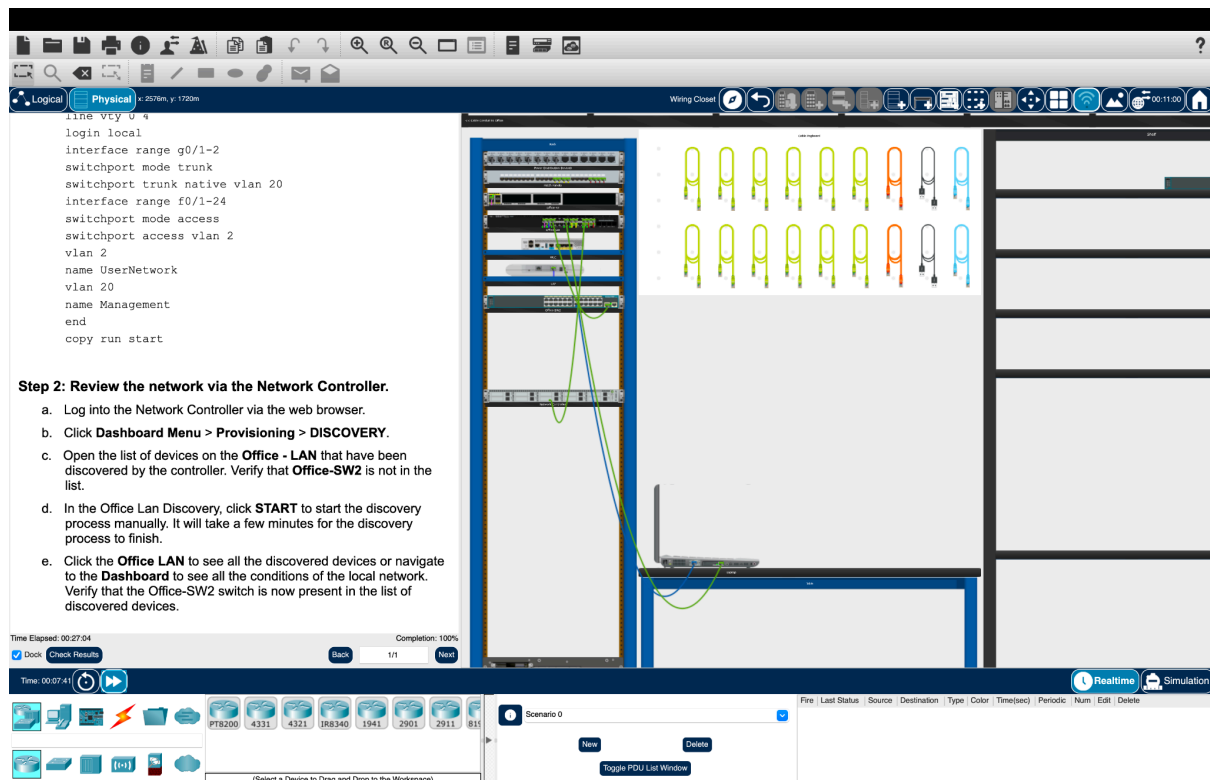
Scenario 0

(Select a Device to Drag and Drop to the Workspace)

11

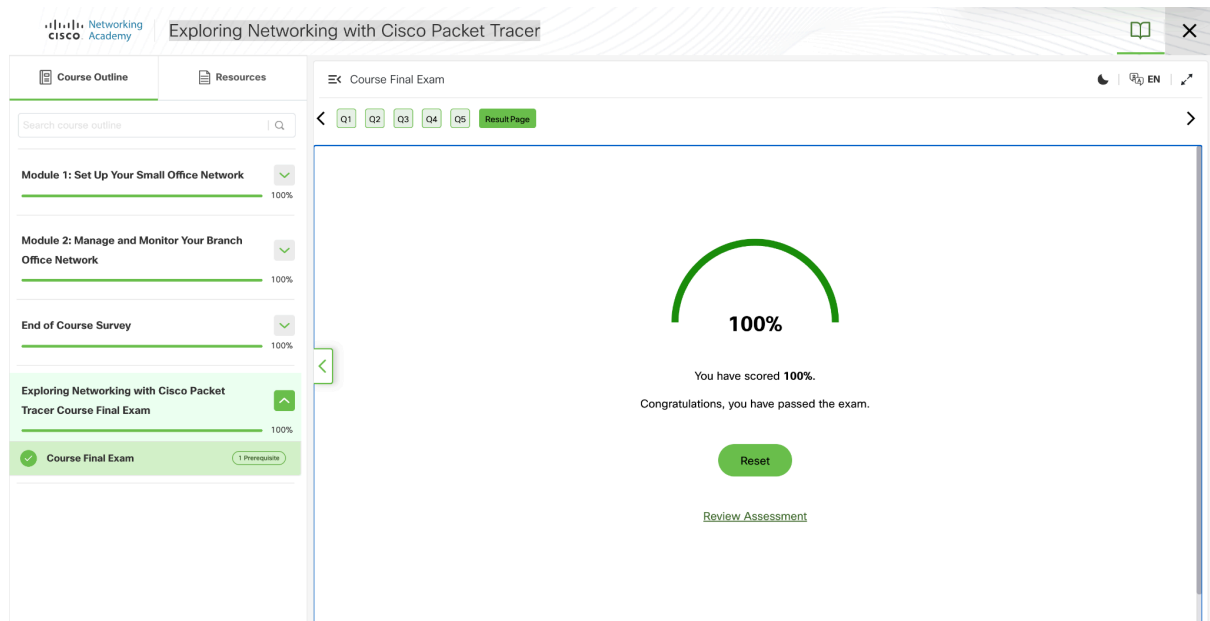
¹⁰ Screenshot of 2.0.7 Packet Tracer - Edit Topologies

¹¹ Screenshot of 2.1.3 Packet Tracer - Monitor Your Network Using a Network Controller



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• Screenshot showing quiz/assessment completion



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¹² Screenshot of 2.1.5 Packet Tracer - Manage and Configure your Network using a Network Controller

¹³ Screenshot showing quiz/assessment completion

Activity 2:

Official certificate:



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Activity 3:

1. What is the main objective of the course “Exploring Networking with Cisco Packet Tracer”?

It provides professional guidance to let us use the Cisco Packet Tracer simulation tool (like wireless devices to connect each other). With these guidelines, it makes us more convenient and easy to use and monitor how a small branch office network works.

2. In the course, what are the main components of the Packet Tracer interface? Briefly explain each one.

¹⁴ Official certificate

Basically, this course focuses on Physical Layer, which includes physical workspace, Menu Bar, Main Toolbar, Common Tools Bar, Device-Type Selection Box, Device-Specific Selection Box, and the Realtime/Simulation Bar.

Menu Bar: Providing function buttons like File, Edit, Options, View, Tools, Help.

Main Tool Bar: Shortcuts for basic tasks like saving, opening, printing, and zooming.

Common Tools Bar: Tools for selecting, deleting, inspecting, drawing, and adding simple/complex PDUs.

Network Component Box: Selecting device types such as Routers, Switches, End Devices, Connections.

Device-Specific Selection Box: Displaying specific models for the chosen device.

Realtime/Simulation Bar: Allowing toggling between Realtime mode and Simulation mode when we are doing packet tracing.

3. What is the purpose of the Device-Type Selection box in Packet Tracer?

The main purpose is used to categorize and select devices, such as routers, switches, end devices, and cables.

4. Explain how to add end devices and network devices in the Logical Workspace.

First of all, we need to put a device in the workspace, then select a device type by choosing it in the Device-Type Selection box. Then, select a device model by clicking on it in the Device-Specific Selection box. Finally, we need to click on a location in the workspace to put our device there.

5. What steps are required to connect two devices using a cable in Packet Tracer?

First of all, we need to place two devices on the workspace, then select the Connections icon which is in the shape of a lightning bolt. After that, select the cable type according to different types, as the Copper Cross-Over cable can only connect similar devices, while the Straight-Through cable can connect different devices. Lastly, select the first device and then select the interface such as FastEthernet0, and then select the second device to make the connection successful.

6. In the course activities, how do you configure an IP address on a PC? Explain the steps.

Firstly click on the PC and then the Desktop tab, IP Configuration, and manually input the IP address like 192.168.1.10, the subnet mask 255.255.255.0, and the IP address of the router 192.168.1.1 as the default gateway. Secondly, click on the router and then the CLI or Config tab, and input the interface configuration using interface gigabitethernet0/0, the IP address 192.168.1.1 and the subnet mask 255.255.255.0 using IP address 192.168.1.1 255.255.255.0, and ensure the interface is active by using the no shutdown command. In the end, once the devices are configured, we can use the ping command to test in the Command Prompt of the

PC by typing ping 192.168.1.1, and if the ping is successful, the devices are connected.

7. What happens if the IP address and subnet mask are configured incorrectly?

It will cause different devices to not communicate with each other immediately. Because of sending the packets to the wrong destination, it may not be able to connect to the networks outside of its local network.

8. During the course tasks, how is a router configured using the CLI? Mention the basic commands used.

1. Click on the router in Packet Tracer and go to the "CLI" tab. Press Enter to begin.

```
http://www.cisco.com/wwl/export/crypto/tool/stqrg.html
```

```
If you require further assistance please contact us by sending email to  
export@cisco.com.
```

```
Cisco CISC01941/K9 (revision 1.0) with 491520K/32768K bytes of memory.  
Processor board ID FTX152400KS  
2 Gigabit Ethernet interfaces  
DRAM configuration is 64 bits wide with parity disabled.  
255K bytes of non-volatile configuration memory.  
249856K bytes of ATA System CompactFlash 0 (Read/Write)
```

```
--- System Configuration Dialog ---
```

```
Would you like to enter the initial configuration dialog? [yes/no]: yes
```

```
At any point you may enter a question mark '?' for help.  
Use ctrl-c to abort configuration dialog at any prompt.  
Default settings are in square brackets '[]'.
```

```
Basic management setup configures only enough connectivity  
for management of the system, extended setup will ask you  
to configure each interface on the system
```

```
Would you like to enter basic management setup? [yes/no]: yes  
Configuring global parameters:
```

```
Enter host name [Router]: 67
```

```
The enable secret is a password used to protect access to  
privileged EXEC and configuration modes. This password, after  
entered, becomes encrypted in the configuration.
```

```
Enter enable secret:
```

```
% No defaulting allowed
```

```
Enter enable secret:
```

```
Press RETURN to get started!
```

2. enable
3. configure terminal
4. interface GigabitEthernet 0/0
5. ip address 192.168.1.1 255.255.255.0
6. no shutdown

```

Router>enable
Router# configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#configure terminal
      ^
% Invalid input detected at '^' marker.

Router(config)#interface GigabitEthernet 0/0
Router(config-if)#ip address 192.168.1.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-3-UPDOWN: Interface GigabitEthernet0/0, changed state to down

7. exit
Router(config-if)#
%LINK-3-UPDOWN: Interface GigabitEthernet0/0, changed state to down
exit
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#exit

```

8. enable
9. copy running-config startup-config
10. Press "Enter" to accept the default filename

```

Router#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]

```

11. show ip interface brief

```

Router#show ip interface brief

```

Interface	IP-Address	OK?	Method	Status	Protocol
GigabitEthernet0/0	192.168.1.1	YES	manual	down	down
GigabitEthernet0/1	unassigned	YES	unset	administratively down	down
Vlan1	unassigned	YES	unset	administratively down	down

```

Router#|

```

12. exit

9. In Simulation Mode, how can you observe the movement of packets?

Firstly, click the stopwatch icon in the bottom right corner to switch from Realtime mode to Simulation mode.

Then, click Add Simple PDU button and click a source device and then a destination device.

In the end, click the Auto Capture / Play button to watch how packets move in real-time

10. What common troubleshooting steps were demonstrated in the course when communication fails?

First, check the IP configuration. If an IP address was wrong, it is recommended to give it a new one.

Try the Ping command to test connections with remote servers.

Next, check DNS to see if the results are "Request timed out" or "No response from server." If they are, the problem could be with the DNS server for the destination.

11. Explain the difference between configuring a device using the Desktop tab and using the CLI tab in Packet Tracer. When would you use each one?

The desktop tab is also called GUI(Graphical User Interface), which is a visual interface with a menu to set IP addresses, interface statuses (On/Off), and routing protocols.

The CLI tab is the Command Line Interface, which is good for professionals, it is based on the command line to control everything that simulates a real Cisco router/switch terminal, requiring commands like enable, ping or other commands as shown before.

When I want to set up a simple topology or network quickly, I will prefer to use the GUI (Desktop tab).

But if I want to do some advanced configurations, such as show the details of IP config, I would like to use the CLI.

12. What is the purpose of the show ip interface brief command in a router? What information does it display?

It gives a quick summary of all the physical interface statuses such as IP address assignments, and Layer 1/Layer 2 connections.

13. Write the basic CLI commands used to:

i. Enter privileged EXEC mode

"enable" to make Router> turns into Router#

ii. Enter global configuration mode

"configure terminal" to make Router# to Router(config)#

iii. Configure an interface IP address

"interface [type/number]" (e.g. "interface FastEthernet0/0")

"ip address [ip address number] [subnet mask number]"(e.g. "ip address 192.168.1.1 255.255.255.0")

"no shutdown" to make it awaken all the time

"end"

All above commands are used to make Router(config)# into Router(config-if)#

iv. Save the configuration

"copy running-config startup-config" to back to Router#

14. During troubleshooting in Packet Tracer, how can you identify whether the problem is caused by Incorrect IP configuration, Wrong cable type or Interface shutdown state?

Issue	Link Light Color	Logical Status
Incorrect IP configuration	Green	Ping Fails
Wrong cable type	Red	Physical layer failure
Interface shutdown state	Red	Manually shutdown